Operator-Regularised Dupire Local-Volatility Calibration

Spatial Penalties, Stability and Recovery

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October 2026

1 Abstract

Recovering local volatility from option prices is an ill-posed inverse problem: distinct surfaces can produce similar prices, and small quote errors can induce large changes in the recovered surface. This project studies whether a spatially varying, Schrödinger-type quadratic penalty improves calibration relative to an SSVI reference and conventional smoothness penalties.

The model estimates a bounded log-variance correction to a training-fitted SSVI surface. A forward Dupire solver supplies prices and discrete tangent sensitivities. The penalty combines first-derivative energies, a positive mass term and nonnegative potentials in sparse-quote regions and strike wings. A fixed-linearisation argument establishes conditional data-to-correction Lipschitz continuity. It does not prove global nonlinear stability or universal accuracy improvement.

The controlled comparison uses a 155-quote Euro Stoxx benchmark, 324 author-supplied synthetic Bates observations and ten newly generated noise draws with known local-volatility truth. Fifty outer folds separate quote interpolation, whole-expiry holdout and known-truth recovery. References, potentials and regularisation strengths are fitted within training partitions.

The full operator reduces Euro Stoxx quote-holdout price RMSE by 34.18% relative to SSVI, but is only 0.93% better than ordinary H1 on that quote split and increases whole-expiry error by 46.81% versus SSVI. Bates price errors improve substantially over SSVI, although H1 is better on whole-expiry holdout. On known-truth recovery, the full operator wins in five of ten draws but its mean full-domain volatility RMSE is 0.09679, versus 0.04096 for SSVI. Unstable draws dominate the mean.

These results support a reproducible inverse-problem investigation, rather than a claim of a superior calibration method. The report separates pricing accuracy, surface recovery and conditional stability, and records unresolved optimisation failures: 72 of 250 selected finite fits reach the iteration limit. The research decision is to resolve optimisation and recovery instability before attributing an advantage to the added potentials. All 61 tests passed; saved fits and reporting evidence were independently checked for this report. Full nested calibration was not rerun.

2 Introduction

An option-pricing model can fit observed quotes while giving an unreliable local-volatility surface. Prices aggregate information through a parabolic equation; inversion attempts to recover a coefficient from those smoothed observations. Sparse maturities, poorly observed wings and quote noise make the inversion sensitive to modelling assumptions.

Dupire's forward equation relates the call-price surface to local variance. Directly recovering variance from maturity derivatives and strike curvature is fragile when only a finite set of observations is available. A PDE-constrained fit instead proposes a surface, prices the observations and adjusts the surface to reduce discrepancies. Regularisation controls what happens between observations and in weakly identified directions.

This project asks whether where a correction is penalised matters beyond ordinary smoothness. A Schrödinger energy motivates combining derivative penalties with a location-dependent potential. The derivative terms discourage variation; the potential discourages the correction's amplitude in selected regions. Sparse quote support and the strike wings supply interpretable locations for stronger shrinkage toward a reference surface.

The analogy is structural. The computation is a classical inverse problem, and a quantum-mechanical interpretation supplies no automatic statistical advantage. Related local-volatility work already uses Tikhonov and Sobolev regularisation [5]. The relevant contribution here is an explicit operator construction, a conditional stability derivation and a controlled empirical comparison. Novelty over the literature is not established by changing terminology.

Three questions guide the evaluation. Does the fitted surface price withheld options better? Does it recover a known local-volatility truth more accurately? What sensitivity guarantee can actually be proved? These questions have different answers in the current experiment. In particular, improved price interpolation does not establish improved surface identification or hedging performance.

3 Data and Evaluation Design

3.1 Public inputs and their scope

The market benchmark contains 155 nonmissing Euro Stoxx implied-volatility quotes at 12 maturities, dated 1 March 2010. The input is a public QuantLib transcription of the Andreasen–Huge example, also used in later literature. It is not a verified byte-for-byte copy of another author's original calibration file. Source zeros denote missing observations. Rates and dividends are zero under the benchmark convention; historical bid–ask spreads are unavailable [2, 3].

The Bates benchmark contains 324 observations at 18 maturities from the unchanged official supplement of Ackerer, Tagasovska and Vatter [6]. It is generated by a jump model. Its synthetic dates are not historical quote dates, and it supplies option prices rather than a known local-volatility truth.

Known-truth experiments regenerate the local-volatility example described by Albani and co-authors [5]. There are 155 quote locations: five maturities from 0.1 to 0.5 years and 31 log-moneyness values from -0.75 to 0.75. Spot is fixed at one. Ten new noise seeds, $20261022 + j$ for $j = 0, \ldots, 9$, replace the authors' unavailable original random realisation. The noise standard deviation is $\max(0.01 C_i^{\text{clean}}, 10^{-6})$. This is not their spot-adjustment experiment.

3.2 Outer and inner partitions

Quote holdout uses five folds, interleaving sorted strikes within each maturity. It tests interpolation among observed expiries. Whole-expiry holdout removes one complete maturity at a time: 12 Euro Stoxx folds and 18 Bates folds. These test reconstruction at missing maturities, including endpoint cases. Ten known-truth noise draws bring the total to 50 outer folds.

Within each outer training set, a four-way interleaved assignment reserves one group for validation. SSVI references and density potentials are rebuilt from the applicable fitting subset. Each penalty selects its own regularisation strength before refitting on the outer training set. Outer prices, implied volatilities and scale values are excluded from fitting and tuning. Requested observation locations are known to the pricing grid; their target values are not used to fit it.

3.3 Interpretation of the experiment

The folds reuse a single market cross-section or a single synthetic surface family. They do not constitute independent market regimes or a future chronological test. Known-truth recovery is measured on a separate dense surface grid of 25 maturities by 121 strikes. Only the regenerated experiment supports that recovery metric.

4 Forward Model and Surface Parameterisation

4.1 Normalised Dupire equation

Let $D(T)$ be the discount factor, $F(T)$ the forward and $x = \log(K/F(T))$. For the normalised call price $c = C/[D(T)F(T)]$, the forward equation is

$$\partial_T c = \frac{1}{2}\sigma^2_{\text{loc}}(x,T)\left(\partial_{xx}c - \partial_x c\right), \qquad c(x,0) = (1-e^x)_+. \tag{1}$$

The solver discretises in normalised strike $k = e^x$ on a grid generated by clustered log-moneyness. This preserves the linear intrinsic payoff in strike. Fixed boundary payoff values close the finite computational domain; boundary conditions are part of the numerical approximation.

The calibration mesh uses 401 spatial nodes and 240 base time steps, augmented by quote expiries. Two initial time intervals use backward-Euler half steps to damp the payoff kink; subsequent steps use Crank–Nicolson. Evaluation reprices on 801 nodes and 480 base time steps. A separate domain check extends the log-moneyness limit from 3 to 4. Discrete tangent equations propagate the Jacobian of prices with respect to correction parameters. Tests compare these sensitivities with centred finite differences.

4.2 Reference plus bounded correction

A price-fitted SSVI surface provides the reference, with conservative sufficient butterfly constraints [4]. Its associated local variance defines $u_{\text{ref}} = \log \sigma^2_{\text{ref}}$. The fitted field is

$$u(x,T) = u_{\text{ref}}(x,T) + (Qh)(x,T), \qquad \sigma_{\text{loc}}(x,T) = \exp\{u(x,T)/2\}. \tag{2}$$

Here $h \in [-2.5, 2.5]^p$ is a latent correction grid with $7 \times 15 = 105$ parameters. The map Q bilinearly interpolates the grid and applies smooth maturity and strike tapers. Outside correction support, the model returns to the reference. Exponentiation ensures positive volatility; it does not guarantee an accurate surface.

The penalty acts on h, rather than directly on the tapered field Qh. This distinction matters: the penalty's geometry, the basis approximation and the actual surface target are separate. A fixed reference and a low-dimensional correction reduce flexibility, but may introduce bias where the true correction lies outside the supported space.

5 Operator Construction and Conditional Theory

5.1 A spatially varying quadratic energy

On nondimensional log-moneyness and maturity coordinates (ξ, s), define

$$\mathcal{E}_V(h) = \int_\Omega \left[\alpha_x |\partial_\xi h|^2 + \alpha_T |\partial_s h|^2 + (\beta + V)h^2\right] d\xi\, ds. \tag{3}$$

With integration-by-parts boundary terms removed, the associated differential operator has the Schrödinger form

$$\mathcal{H} = -\alpha_x \partial_{\xi\xi} - \alpha_T \partial_{ss} + \beta + V. \tag{4}$$

The benchmark uses $\alpha_x = 0.005$, $\alpha_T = 0.002$ and $\beta = 10^{-4}$ before trace normalisation. A Gaussian kernel density from fitting-quote locations is normalised to confidence $q \in [0, 1]$. Its potential is $V_{\text{density}} = 0.2(1 - q)$. Kernel bandwidths are 0.12 times each coordinate range. This is a support-location proxy, not an estimated noise covariance.

A quadratic wing potential rises from zero beyond $0.44 \max |x|$ to 0.5 at the outermost correction nodes. The full penalty uses both potentials, so $0 \le V \le 0.7$. Greater potential shrinks the correction toward the reference; it does not force volatility toward zero.

Positive nodal and edge quadrature give the implemented matrix

$$\begin{aligned} R_{\text{raw}} &= \beta M + \alpha_x D_\xi^{\mathsf{T}} M_\xi D_\xi + \alpha_T D_s^{\mathsf{T}} M_s D_s + \text{diag}(m_i V_i), \\ R &= R_{\text{raw}} / \{\text{tr}(R_{\text{raw}})/p\}. \end{aligned} \tag{5}$$

The positive mass term makes R positive definite. If $R = U\,\text{diag}(r_j)U^{\mathsf{T}}$, then $h^{\mathsf{T}}Rh = \sum_j r_j (U^{\mathsf{T}}h)_j^2$: high-energy modes receive stronger penalties. This spectral interpretation motivates the construction. Each method is trace-normalised and tuned separately, so adding a potential does not imply a larger fitted penalty in matrix order.

5.2 What the Lipschitz bound establishes

Freeze the reference, observation scales, interpolation map, Jacobian and penalty. Write the standardised linear experiment as $b = Ah_* + \epsilon$, let $\tau = n\lambda/p > 0$, and define

$$G = A^{\mathsf{T}}A + \tau R, \qquad \widehat{h}(b) = G^{-1}A^{\mathsf{T}}b. \tag{6}$$

For $\mu = \lambda_{\min}(R) > 0$, the unique unconstrained linearised estimate satisfies

$$\|\widehat{h}(b + \Delta b) - \widehat{h}(b)\|_2 \le \frac{\|A\|_2}{\tau\mu} \|\Delta b\|_2. \tag{7}$$

Indeed, the difference equals $G^{-1}A^{\mathsf{T}}\Delta b$ and $\lambda_{\min}(G) \ge \tau\mu$. The exact Lipschitz constant is $\|G^{-1}A^{\mathsf{T}}\|_2$; the displayed value is an upper bound. Unweighted price perturbations add a factor $\|W\|_2$ because $\Delta b = W\Delta C$.

The conclusion is conditional data-to-correction continuity for a fixed, unconstrained linearisation. It is not a global guarantee for the refitted, tuned, box-constrained nonlinear pipeline. A continuous finite PDE objective on the compact parameter box attains a minimum, but this does not establish uniqueness or convergence to a global optimum.

5.3 Smoothness class and estimation risk

Let E_0 denote the same continuum energy with $V = 0$. A bounded nonnegative potential satisfies

$$E_0(h) \le E_V(h) \le (1 + V_{\max}/\beta)E_0(h). \tag{8}$$

The norms are equivalent for fixed positive coefficients: the potential changes spatial shrinkage within the same anisotropic H1 function class. It does not introduce a higher Sobolev order or imply global spatial Lipschitz regularity of a two-dimensional surface.

Even in the fixed linear experiment, stronger shrinkage trades variance for bias. If $K = G^{-1}A^{\mathsf{T}}$, $E\epsilon = 0$ and $\mathrm{Cov}(\epsilon) = \Sigma$, the exact risk for a fixed linear target Bh_* is

$$E\|B\widehat{h} - Bh_*\|^2 = \|B(KA - I)h_*\|^2 + \mathrm{tr}(BK\Sigma K^{\mathsf{T}}B^{\mathsf{T}}). \tag{9}$$

For scalar data $b = h_* + \epsilon$ with unit noise variance, ridge risk is $(t^2h_*^2 + 1)/(1 + t)^2$. Increasing t from one to two helps exactly when $h_*^2 < 5/7$. This counterexample rules out an assumption-free accuracy advantage from extra shrinkage. An operator aligned with the true prior precision has a Bayesian justification in a correctly specified linear Gaussian model; quote density alone does not establish that alignment.

6 Calibration and Controlled Comparisons

6.1 Objective and numerical updates

Let $\mathcal{P}(h)$ be the vector of PDE prices and y the fitting prices. The objective is

$$\min_{h\in[-2.5,2.5]^p} \frac{1}{n}\|W(\mathcal{P}(h)-y)\|_2^2 + \frac{\lambda}{p}h^{\mathsf{T}}Rh. \tag{10}$$

For market and Bates quotes, $W_{ii} = 1/\max(0.005\,\text{Vega}_i, 10^{-5})$. This declared scale weights errors by an approximate half-volatility-point price change. It is not measured bid–ask uncertainty. The known-truth experiment uses its declared noise scales.

With standardised residual r_k and Jacobian A_k, a damped Gauss–Newton step solves

$$\left(\frac{A_k^{\mathsf{T}}A_k}{n} + \frac{\lambda}{p}R + \eta I\right)\delta = -\left(\frac{A_k^{\mathsf{T}}r_k}{n} + \frac{\lambda}{p}Rh_k\right). \tag{11}$$

Backtracking seeks objective decrease and projection enforces bounds. The initial budget is 50 iterations. Selected fits that exhaust it receive training-only L-BFGS-B polishing at their already selected regularisation strength. This improves the optimisation attempt without reselecting on outer outcomes; its success must still be checked.

6.2 Comparator and selection design

The comparison includes the SSVI PDE reference, unregularised correction, H1, H2, H1 plus density, H1 plus wing, and the full operator. Direct SSVI prices provide a numerical reference check. All correction methods share the reference, PDE solver, grid, taper, coordinates, scale convention and partitions. H2 uses second-difference penalties in these same latent coordinates. These are controlled adaptations, not exact reproductions of each cited paper's volatility-space estimator.

The initial regularisation grid is $10^{-4}, 10^{-3}, \ldots, 10^{2}$. A deterministic extension tests two more values when a finite boundary value is preferred. Returning the unchanged reference, represented by $\lambda = \infty$, is also a candidate. Selection minimises inner-validation scaled price error. The audited benchmark uses this nested selection procedure, rather than a generalised cross-validation claim from earlier project versions.

7 Held-Out Pricing Results

7.1 Absolute errors and relative comparisons

Price RMSE below pools the held-out quote errors for each protocol. Values are basis points of spot: one unit in the table equals 10^{-4} in normalised price. These are pricing-error units, not volatility points or investment returns.

Table 1: Held-out normalised price RMSE, multiplied by 10,000.

Dataset / holdout	SSVI	H1	H2	Full
Euro Stoxx / quote holdout	9.94	6.61	6.23	6.55
Euro Stoxx / expiry holdout	20.54	31.14	24.50	30.16
Bates / quote holdout	16.70	4.27	4.96	4.13
Bates / expiry holdout	15.90	3.57	5.08	3.90

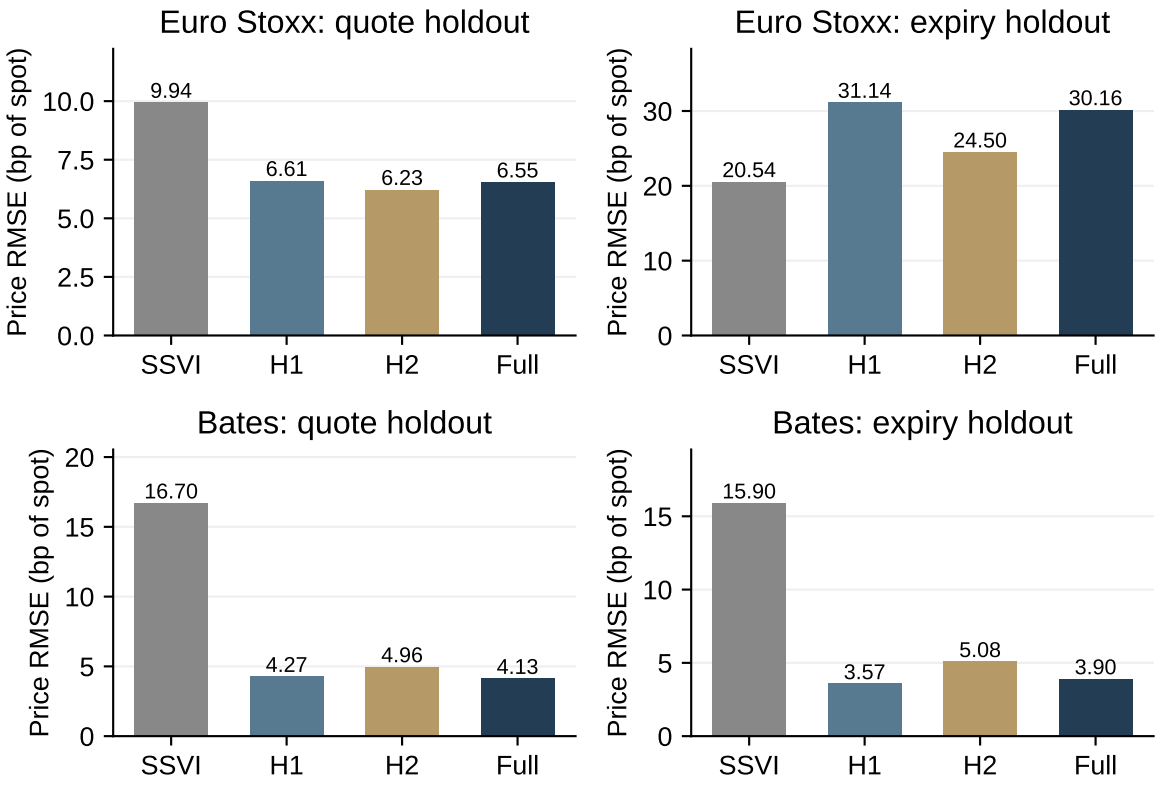


Figure 1: Absolute held-out price errors. The full operator improves substantially on SSVI in three protocols, but not Euro Stoxx whole-expiry holdout. The best conventional comparator varies by setting.

Euro Stoxx quote interpolation benefits from corrections, but the full potential is not the best correction method. H2 has a lower absolute error. On whole-expiry holdout, the SSVI reference has the lowest error of these four methods. This change in ranking is evidence that interpolation among expiries and reconstructing a missing expiry are materially different tasks.

7.2 Magnitude and uncertainty of the changes

Define percentage change as $100(\text{RMSE}_{\text{full}}/\text{RMSE}_{\text{comparator}} - 1)$. Negative values favour the full operator.

Table 2: Full-operator price RMSE change relative to each comparator.

Dataset / holdout	vs SSVI	vs H1	vs H2
Euro Stoxx / quote holdout	-34.18%	-0.93%	+5.12%
Euro Stoxx / expiry holdout	+46.81%	-3.17%	+23.07%
Bates / quote holdout	-75.30%	-3.34%	-16.89%
Bates / expiry holdout	-75.44%	+9.39%	-23.12%

The full operator lowers Bates quote-holdout error by 75.30% relative to SSVI and by 3.34% relative to H1. For missing Bates expiries, it still improves by 75.44% over SSVI, while the full operator's RMSE is 9.39% higher than H1's. Large gains over the reference therefore do not establish an advantage from the added potentials.

The Euro Stoxx quote-holdout ablations give 6.52 basis points for H1 plus density and 6.66 for H1 plus wing, versus 6.55 for full and 6.61 for H1. Combining both potentials is therefore not uniformly helpful.

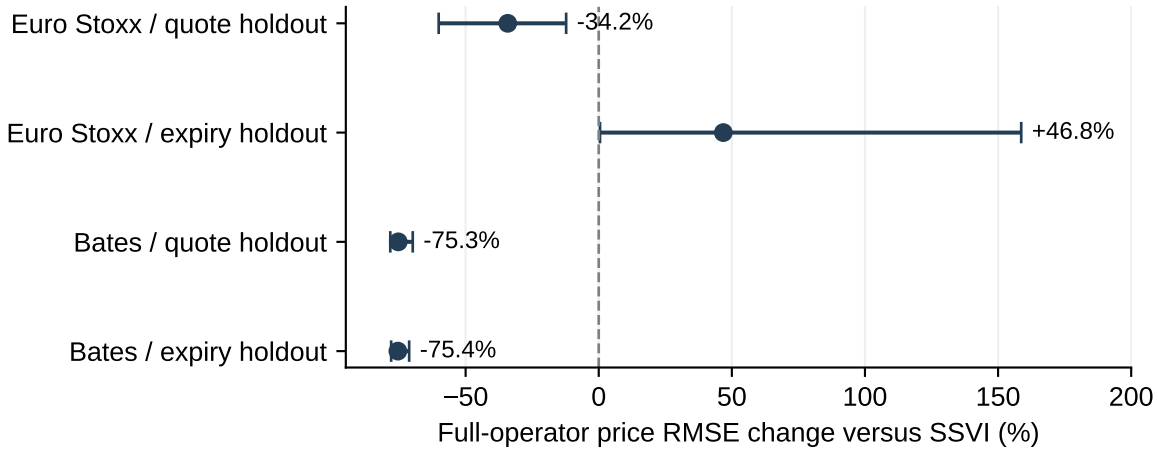


Figure 2: Full operator versus SSVI: point changes and descriptive 95% expiry-cluster bootstrap intervals, using 5,000 resamples. The horizontal scale permits the wide Euro Stoxx expiry interval to remain visible.

The Euro Stoxx quote-holdout interval is approximately $[-60.12\%, -12.24\%]$; its whole-expiry interval is $[0.48\%, 158.69\%]$. Intervals for the full operator versus H1 include zero in both Euro Stoxx protocols. These resample 12 Euro Stoxx or 18 Bates maturity clusters within a fixed cross-section. They are descriptive and do not establish generalisation across market regimes, account for all research selection, or prove a universal ranking.

8 Recovery of Known Local Volatility

8.1 All ten noise draws

Only the regenerated Albani experiment has a supplied local-volatility truth. Full-domain recovery evaluates σ_{loc} directly on the dense grid, rather than inferring accuracy from price residuals. Volatility is expressed as a decimal annualised level. For example, an RMSE of 0.04 is four volatility percentage points.

Table 3: Full-domain local-volatility RMSE across ten noise draws.

Method	Mean	Median	Worst draw
SSVI reference	0.04096	0.04100	0.04608
H1	0.09546	0.03673	0.38506
H1 + density	0.09662	0.03838	0.38395
H1 + wing	0.10024	0.03832	0.38373
H2	0.11643	0.07418	0.37632
Full operator	0.09679	0.03868	0.38304
Unregularised	0.19237	0.17153	0.36514

The full operator beats SSVI in five draws. Its median is slightly lower, but the mean is 136.29% higher. The central-region mean, restricting $|x| \leq 0.35$, is also worse: 0.04071 versus 0.02221. The adverse mean is therefore not confined to distant wings. H1 and the full operator both show large draw-to-draw variability; H2 also fails to improve mean recovery.

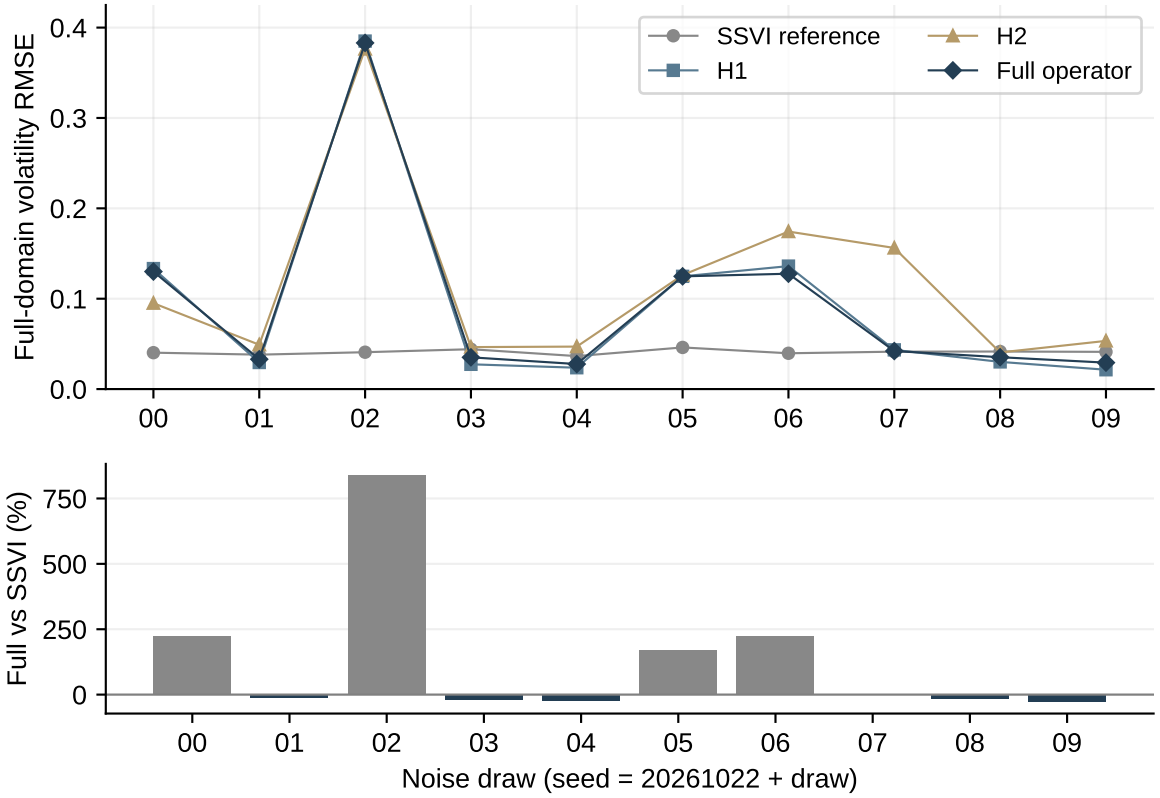


Figure 3: Every noise draw is retained. Upper panel: full-domain recovery errors. Lower panel: full-operator error change relative to SSVI. The large error on draw 02 is visible rather than removed as an outlier.

8.2 What the fitted curves reveal

For illustration, Figure 4 shows the first noise draw in seed order, rather than selecting the best result. The known truth is

$$\sigma_*(x,T) = \begin{cases} 0.4 - 0.16e^{-T/2}\cos(4\pi x/5), & |x| \le 0.4, \\ 0.4, & |x| > 0.4. \end{cases} \tag{12}$$

It follows the published synthetic specification, including its piecewise transition. The clean prices are generated with a separate, finer solver: 1,601 spatial nodes and 1,200 time steps, checked against 2,401 and 1,800. This reduces shared-grid inverse-crime risk, although both solvers still implement the same model family.

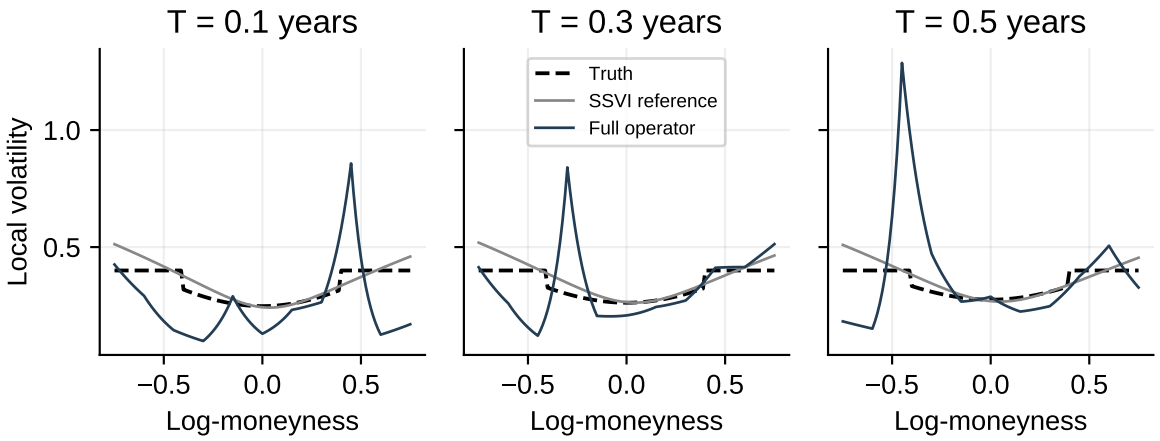


Figure 4: Local-volatility slices for noise draw 00 at 0.1, 0.3 and 0.5 years. Dashed curves are known truth. The SSVI reference and full operator are displayed without smoothing or rescaling their fitted values. The all-draw analysis, rather than these illustrative slices, determines the recovery conclusion.

The PDE smooths information, and the finite quote grid does not identify every variation of the surface equally well. A correction can improve price fit while moving the latent surface away from truth. Regularisation may reduce this instability, but the current choice of potential and its price-based tuning do not demonstrate reliable recovery gains.

The failure cannot be assigned entirely to the statistical penalty. Many selected fits are incompletely optimised, the latent basis restricts representable corrections, and the reference can be misspecified. These mechanisms require separate experiments. The current evidence establishes a failure of the implemented pipeline to deliver consistent recovery, without isolating a single cause.

9 Numerical and Optimisation Diagnostics

9.1 Discretisation and price-shape checks

All 50 completed folds have recorded diagnostics. The largest calibration-to-refined-grid price RMSE is 8.68×10^{-6} in normalised price, or 0.0868 basis points of spot. Expanding the domain changes prices by at most 1.71×10^{-6} RMSE, or 0.0171 basis points. These errors are small relative to the reported aggregate pricing errors; they do not make the inversion itself well posed.

The recorded diagnostic grids show zero strike-convexity and calendar-monotonicity violations. Minimum time value is -1.78×10^{-15}, consistent with floating-point noise. These are finite-grid checks, not a proof of global static arbitrage freedom on every interpolated strike and maturity.

9.2 Incomplete calibration is a material limitation

Table 4: Diagnostics for the 250 selected finite-correction fits.

Recorded condition	Count
At initial iteration budget	72 / 250
At least one active amplitude bound	117 / 250
Selected finite lambda at search boundary	11 / 250
Training-only polishing attempts	72
Polishing optimiser reports success	3 / 72

The 250 fits cover the five tuned penalty variants across 50 folds; the unregularised and unchanged-reference comparators are additional. An active bound may be a valid constrained optimum. An iteration-limit stop does not establish one. Only three of the 72 polishing attempts report optimiser success, so polishing cannot be described as resolving convergence.

A fair model comparison requires comparable solution quality, not merely the same iteration allowance. These diagnostics limit attribution of the observed ranking to the regulariser alone. They also make a claim of production readiness premature. The results are retained as the output of the recorded finite-budget implementation, with this qualification attached.

10 Limitations and Next Experiments

The strongest empirical limitation is inconsistent recovery of known truth. The strongest numerical limitation is unresolved optimisation. A useful next step is to freeze the existing inputs, splits and selected regularisation strengths, then rerun the failed fits with a larger budget, multiple initialisations and recorded projected-gradient norms. Objective values, active constraints and recovered surfaces should be compared before retuning anything. This would separate solver failure from the statistical behaviour of the selected penalty.

A subsequent study should distinguish price-optimal tuning from recovery-optimal tuning in simulation. Recovery-optimal tuning can be used as an oracle diagnostic, clearly labelled as unavailable on market data. Comparing both reveals whether failure comes from the correction family, optimisation, or a price objective that weakly identifies the target surface.

Support-based potentials also need direct stress tests. Low density is not the same as high noise or reference accuracy. A deliberately misspecified reference in sparse regions can test whether stronger shrinkage preserves the wrong surface. Potential ablations, more noise draws and alternate truths should be predeclared. A limited finer-parameter-grid diagnostic already exists in the repository; it is not broad evidence of robustness to model resolution.

The market analysis needs multiple dates and reliable bid–ask data before external usefulness can be assessed. No experiment here measures hedging P&L, quote execution, transaction costs or real-time operational behaviour. The Bates benchmark is synthetic and cannot substitute for historical stress regimes.

For a stronger theorem, the research must specify a truth class, observation-noise model, target norm and linked limits for grid error and regularisation strength. An improved convergence rate would then need assumptions under which this operator aligns more favourably with truth than H1 or H2. The present finite-dimensional stability bound is a narrower result. These experiments and theorem extensions are proposed work, not completed results.

11 Conclusion

The project builds a complete local-volatility calibration pipeline: a training-fitted SSVI reference, bounded log-variance correction, differentiable Dupire pricing, spatial operator penalties and nested comparisons. The Schrödinger form gives an interpretable combination of derivative energy and location-dependent shrinkage. Positive definiteness supplies a clear conditional stability statement for a fixed linearisation.

The empirical result is mixed. The full operator improves price interpolation relative to SSVI and performs strongly on the Bates price benchmark. It does not dominate conventional H1 and H2 penalties. It worsens Euro Stoxx whole-expiry prediction relative to SSVI, and fails to improve average recovery of known local volatility across ten noise draws.

The defensible conclusion is that the proposed spatial penalty is an investigable modelling choice, rather than an established superior estimator. Its value must be demonstrated against strong smoothness comparators under reliable optimisation and a precisely defined target. The current contribution includes making that distinction measurable and preserving the negative evidence alongside the gains.

12 Reproducibility and Sources

12.1 Version and evidence checked for this report

The report uses repository commit `b1ba3bcf7eee` on the main branch [1]. The source was checked out afresh. The calibration archive and all 644 manifest-listed files were verified against SHA-256 fingerprints. The reporting analysis was rerun; nine saved tables or summaries reproduced numerically.

Headline comparisons were recomputed from held-out predictions. All 140 known-truth method/region checks were recomputed from recovery arrays. The 50 saved folds were restored and repriced on the recorded refined PDE grid; all 400 method/fold prediction checks matched within 10^{-16} in normalised price. Exact differences are retained in the companion verification JSON. All 61 repository tests passed. This verifies the supplied evidence and numerical replay. It does not constitute a fresh rerun of the complete nested optimisation and hyperparameter search.

The numerical experiment and replay entry points are:

```
scripts/23_literature_benchmarks.py

scripts/24_literature_diagnostics.py

scripts/26_analyze_literature_results.py
```

The mathematical construction is documented in the repository theory note; input provenance and the controlled-comparison scope are recorded separately. The companion LinkedIn file contains the project entry and media details without inferring project dates.

October 2026 review: the public research decision note records the leading comparison, evidence limits and next experiment. The pinned source versions below identify the experiments; the review changes presentation and the explicitly described code checks.

References

[1] Hussain, F. H. Operator-regularized Dupire. Source, saved calibrations, diagnostics and audit tables. Versioned repository. Report assembled October 2026.

[2] Project input-provenance record. Literature benchmark inputs and source links.

[3] QuantLib. Andreasen–Huge volatility interpolation test suite. Public Euro Stoxx example. Source file.

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[5] Albani, V., Ascher, U., Yang, X. and Zubelli, J. Data driven recovery of local volatility surfaces. arXiv:1512.07660. Synthetic specification regenerated with new noise draws.

[6] Ackerer, D., Tagasovska, N. and Vatter, T. Deep Smoothing of the Implied Volatility Surface. NeurIPS 2020. Paper; official supplement supplying the Bates observations.